KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5341 | 5 Credits: 4 | Program Core | Source-grounded

Big Data Analytics

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5341 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5341.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence & Machine Learning
Course code	5341
Course title	Big Data Analytics
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

22 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	Summarize the concepts of Big Data 15 Understanding Summarize the basic concepts in Big Data	See official syllabus
CO2	14 Understanding Analytics and NoSQL Demonstrate Map Reduce programming in	See official syllabus
CO3	15 Applying Hadoop Environment Make use of basic programming tools like Hive	See official syllabus
CO4	16 Applying and Pig in Hadoop ecosystems	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Big Data concepts	7	As specified
Module 2	Summarize the basic concepts in Big Data Analytics and NoSQL	5	As specified
Module 3	Describe Map Reduce programming in Hadoop Environment	5	As specified
Module 4	Outline the basic concepts of Hive and Pig	5	As specified

MODULE 1

Big Data concepts

This module is presented from the official syllabus scope for Big Data Analytics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Big Data concepts
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

Summarize the Characteristics of Big Data

Summarize the Characteristics of Big Data is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the Characteristics of Big Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the characteristics of big data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the Characteristics of Big Data means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarize the Characteristics of Big Data □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Classify Digital Data 2 Understanding Describe Big data Analytics and its

Classify Digital Data 2 Understanding Describe Big data Analytics and its is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify Digital Data 2 Understanding Describe Big data Analytics and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify digital data 2 understanding describe big data analytics and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classify Digital Data 2 Understanding Describe Big data Analytics and its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Classify Digital Data 2 Understanding Describe Big data Analytics and its □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

2 Understanding applications

2 Understanding applications is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Understanding applications [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compare Big Data and Data Warehouse

Compare Big Data and Data Warehouse is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare Big Data and Data Warehouse using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare big data and data warehouse should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare Big Data and Data Warehouse means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Compare Big Data and Data Warehouse □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Describe Data Mining Concepts

Describe Data Mining Concepts is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe Data Mining Concepts using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe data mining concepts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe Data Mining Concepts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Describe Data Mining Concepts [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss Preprocessing techniques 3 Understanding Describe Data Visualization and List some

Discuss Preprocessing techniques 3 Understanding Describe Data Visualization and List some is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss Preprocessing techniques 3 Understanding Describe Data Visualization and List some using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss preprocessing techniques 3 understanding describe data visualization and list some should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss Preprocessing techniques 3 Understanding Describe Data Visualization and List some means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Discuss Preprocessing techniques 3 Understanding Describe Data Visualization and List some □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding tools used for Visualization Introduction to Bigdata: Data –Big data –Examples of Big data Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics. Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (Basics only) Data Visualization: Definition, Need for Visualization, List some tools used for Visualization

3 Understanding tools used for Visualization Introduction to Bigdata: Data –Big data –Examples of Big data Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics. Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (Basics only) Data Visualization: Definition, Need for Visualization, List some tools used for Visualization is an official topic in Module 1 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding tools used for Visualization Introduction to Bigdata: Data –Big data –Examples of Big data Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics. Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods Data Preprocessing -Need of data preprocessing, data quality, data preprocessing

methods- data cleaning, data integration, data reduction, data transformation (Basics only) Data Visualization: Definition, Need for Visualization, List some tools used for Visualization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding tools used for visualization introduction to bigdata: data –big data –examples of big data characteristics of big data- volume, variety, velocity, and other characteristics. classification /types of f digital data: structured, unstructured and- semi-structured data warehouse: definition, coexistence of big data and data warehouse. data mining- introduction, data mining goals, types of data, steps to discover knowledge from data (stages of the data mining process), data mining techniques, knowledge representation methods data preprocessing -need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (basics only) data visualization: definition, need for visualization, list some tools used for visualization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding tools used for Visualization Introduction to Bigdata: Data –Big data –Examples of Big data Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics. Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (Basics only) Data Visualization: Definition, Need for Visualization, List some tools used for Visualization means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding tools used for Visualization Introduction to Bigdata: Data –Big data –Examples of Big data Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics. Classification /Types of f Digital Data: Structured, Unstructured and- Semi-Structured Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data (Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation (Basics only) Data Visualization: Definition, Need for Visualization, List some tools used for Visualization □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Big Data concepts

    M1.01       Summarize the Characteristics of Big Data      2
Understanding

    M1.02       Classify Digital Data                          2
Understanding

                Describe Big data Analytics and its
    M1.03                                                      2
Understanding
                applications
    M1.04       Compare Big Data and Data Warehouse            1
Understanding
    M1.05       Describe Data Mining Concepts                  2
Understanding
    M1.06       Discuss Preprocessing techniques               3
Understanding
                Describe Data Visualization and List some
    M1.07                                                      3
Understanding
                tools used for Visualization
Contents:
Introduction to Bigdata: Data –Big data –Examples of Big data
Characteristics of Big Data- Volume, Variety, Velocity, and other
characteristics.
Classification /Types of f Digital Data: Structured, Unstructured and- Semi-
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize the basic concepts in Big Data Analytics and NoSQL

This module is presented from the official syllabus scope for Big Data Analytics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the basic concepts in Big Data Analytics and NoSQL
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Analytics and the major challenges faced 3 Understanding by Big Data.

Analytics and the major challenges faced 3 Understanding by Big Data. is an official topic in Module 2 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Analytics and the major challenges faced 3 Understanding by Big Data. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, analytics and the major challenges faced 3 understanding by big data. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Analytics and the major challenges faced 3 Understanding by Big Data. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Analytics and the major challenges faced 3 Understanding by Big Data. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment

Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment is an official topic in Module 2 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the concepts of data science 2 understanding summarize the big data environment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding terminologies

4 Understanding terminologies is an official topic in Module 2 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding terminologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding terminologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding terminologies means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding terminologies [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize the concepts of NoSQL

Summarize the concepts of NoSQL is an official topic in Module 2 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the concepts of NoSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the concepts of nosql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the concepts of NoSQL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Summarize the concepts of NoSQL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline the concepts of NewSQL 1 Understanding Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics Data Science: Definition , Techniques used in Data Science - classification, regression and clustering Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems. NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL

Outline the concepts of NewSQL 1 Understanding Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics Data Science: Definition , Techniques used in Data Science - classification, regression and clustering Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems. NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL is an official topic in Module 2 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the concepts of NewSQL 1 Understanding Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics Data Science: Definition , Techniques used in Data Science - classification, regression and clustering Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and

Distributed Systems. NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the concepts of newsql 1 understanding big data analytics: definition of big data analytics, requirement of high-performance analytics --benefits of big data analytics, classification of analytics, top challenges facing big data, importance of big data analytics, applications of big data analytics data science: definition , techniques used in data science - classification, regression and clustering terminologies used in big data environment: in-memory analytics, in-database processing, symmetric multi-processor systems, massively parallel processing, difference between parallel and distributed systems. nosql: definition, types of nosql databases, use nosql - storing and manipulating data. newsql: characteristics of newsql, comparison of sql, nosql, and newsql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the concepts of NewSQL 1 Understanding Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics Data Science: Definition , Techniques used in Data Science - classification, regression and clustering Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems. NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline the concepts of NewSQL 1 Understanding Big Data Analytics: Definition of Big Data analytics, Requirement of high-performance analytics --Benefits of Big Data analytics, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics Data Science: Definition , Techniques used in Data Science - classification, regression and clustering Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems. NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the basic concepts in Big Data Analytics and NoSQL

                 Explain the importance of Big Data
    M2.01        Analytics and the major challenges faced         3
Understanding
                 by Big Data.
    M2.02        Outline the concepts of Data Science             2
Understanding
                 Summarize the Big Data Environment
    M2.03                                                         4
Understanding
                 terminologies

    M2.04        Summarize the concepts of NoSQL                  4
Understanding

    M2.05        Outline the concepts of NewSQL                   1
Understanding
                 Series Test – I                                  1
Contents:
Big Data Analytics: Definition of Big Data analytics, Requirement of high-
performance
analytics --Benefits of Big Data analytics, Classification of Analytics, Top
Challenges Facing
Big Data, Importance of Big Data Analytics, Applications of Big Data Analytics
Data Science: Definition , Techniques used in Data Science - classification,
regression and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe Map Reduce programming in Hadoop Environment

This module is presented from the official syllabus scope for Big Data Analytics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe Map Reduce programming in Hadoop Environment
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Explain the fundamentals of Hadoop

Explain the fundamentals of Hadoop is an official topic in Module 3 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the fundamentals of Hadoop using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the fundamentals of hadoop should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the fundamentals of Hadoop means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the fundamentals of Hadoop □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Concepts of HDFS

Concepts of HDFS is an official topic in Module 3 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Concepts of HDFS using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, concepts of hdfs should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Concepts of HDFS means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Concepts of HDFS [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and

List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and is an official topic in Module 3 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the components of hadoop ecosystem 3 understanding illustrate the concept of hdfs and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

2 Applying MapReduce

2 Applying MapReduce is an official topic in Module 3 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying MapReduce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying mapreduce should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying MapReduce means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 2 Applying MapReduce [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Use the different function in MapReduce 5 Applying Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting.

Use the different function in MapReduce 5 Applying Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. is an official topic in Module 3 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Use the different function in MapReduce 5 Applying Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, use the different function in mapreduce 5 applying hadoop: features of hadoop, key advantages of hadoop,hadoop distributions, integrated hadoop systems offered by leading market vendors, cloud-based hadoop solutions overview of hadoop ecosystems and hadoop core components, hdfs, processing data with hadoop map reduce programming: introduction, mapper, reducer, combiner, partitioner, searching, sorting. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Use the different function in MapReduce 5 Applying Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Use the different function in MapReduce 5 Applying Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe Map Reduce programming in Hadoop Environment

    M3.01        Explain the fundamentals of Hadoop                 4
Understanding
    M3.02        Concepts of HDFS                                   1
Understanding
    M3.03        List the components of Hadoop Ecosystem            3
Understanding
                 Illustrate the concept of HDFS and
    M3.04                                                           2
Applying
                 MapReduce
    M3.04        Use the different function in MapReduce            5
Applying
Contents:
Hadoop: Features of Hadoop, Key Advantages of Hadoop,Hadoop Distributions,
Integrated
Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions
Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing
Data with Hadoop
Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner,
Searching, Sorting.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Outline the basic concepts of Hive and Pig

This module is presented from the official syllabus scope for Big Data Analytics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the basic concepts of Hive and Pig
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Demonstrate the basic concepts of Hive

Demonstrate the basic concepts of Hive is an official topic in Module 4 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate the basic concepts of Hive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate the basic concepts of hive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate the basic concepts of Hive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Demonstrate the basic concepts of Hive [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Apply Hive Query Language Statements

Apply Hive Query Language Statements is an official topic in Module 4 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply Hive Query Language Statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply hive query language statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply Hive Query Language Statements means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Apply Hive Query Language Statements [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Make use of Hive User Defined Function

Make use of Hive User Defined Function is an official topic in Module 4 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of Hive User Defined Function using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of hive user defined function should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of Hive User Defined Function means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Make use of Hive User Defined Function □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Make use of the basic concepts of Pig

Make use of the basic concepts of Pig is an official topic in Module 4 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of the basic concepts of Pig using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of the basic concepts of pig should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of the basic concepts of Pig means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Make use of the basic concepts of Pig □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Make use of HDFS Commands 3 Applying Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han, Champaign Micheline Kamber,Jian Pei Simon Fraser ‘” Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/

Make use of HDFS Commands 3 Applying Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han, Champaign Micheline Kamber,Jian Pei Simon Fraser ‘” Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ is an official topic in Module 4 of Big Data Analytics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of HDFS Commands 3 Applying Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han, Champaign Micheline Kamber,Jian Pei Simon Fraser ‘” Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, ―Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, ― Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of hdfs commands 3 applying hive: introduction – architecture - data types - file formats - hive query language statements – partitions – bucketing – views - sub- query – joins – aggregations. hive user defined function - serialization and deserialization (serde). pig: introduction - anatomy – features – use case for pig - pig latin overview - pig primitive data types - running pig - execution modes of pig - hdfs commands - user-defined functions. pig versus hive text / reference t/r book title/author t1 t1 seema acharya, subhasini chellappan, "big data analytics", wiley, 2015 jiawei han, champaign micheline kamber,jian pei simon fraser ‘” data mining t2 concepts and techniques third edition” frank j ohlhorst, ―big data and analytics: turning big data into big money, r1 wiley and sas business series, 2012. r2 tom white, ― hadoop: the definitive guide third edition, o‘reily media, 2012 emc education services, data science and big data analytics: discovering, r3 analyzing, visualizing and presenting data. john wiley & sons, 2015. publishers – 2012. online resources sl.no website link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of HDFS Commands 3 Applying Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han, Champaign Micheline Kamber,Jian Pei Simon Fraser ‘” Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-�� Make use of HDFS Commands 3 Applying Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations. Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined Functions. Pig Versus Hive Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han, Champaign Micheline Kamber,Jian Pei Simon Fraser ‘” Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom

White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Outline the basic concepts of Hive and Pig

      M4.01        Demonstrate the basic concepts of Hive          4
Applying
      M4.02        Apply Hive Query Language Statements            3
Applying
      M4.03        Make use of Hive User Defined Function          2
Applying
      M4.04        Make use of the basic concepts of Pig           4
Applying
      M4.05        Make use of HDFS Commands                       3
Applying
                   Series Test -2                                  1
Contents:
Hive: Introduction – Architecture - Data Types - File Formats - Hive Query
Language
Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations.
Hive User
Defined Function - Serialization and Deserialization (SERDE).
Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview -
Pig Primitive
Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - User-Defined
Functions. Pig Versus Hive

Text / Reference

T/R                                        Book Title/Author
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Summarize the Characteristics of Big Data. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the Characteristics of Big Data*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Classify Digital Data 2 Understanding Describe Big data Analytics and its. (3 marks)

Answer: Begin with the standard definition or identity of *Classify Digital Data 2 Understanding Describe Big data Analytics and its*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding applications. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Compare Big Data and Data Warehouse. (3 marks)

Answer: Begin with the standard definition or identity of *Compare Big Data and Data Warehouse*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Analytics and the major challenges faced 3 Understanding by Big Data.. (3 marks)

Answer: Begin with the standard definition or identity of *Analytics and the major challenges faced 3 Understanding by Big Data.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the concepts of Data Science 2 Understanding Summarize the Big Data Environment*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding terminologies. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding terminologies*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize the concepts of NoSQL. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize the concepts of NoSQL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Explain the fundamentals of Hadoop. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the fundamentals of Hadoop*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Concepts of HDFS. (3 marks)

Answer: Begin with the standard definition or identity of *Concepts of HDFS*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and. (3 marks)

Answer: Begin with the standard definition or identity of *List the components of Hadoop Ecosystem 3 Understanding Illustrate the concept of HDFS and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Applying MapReduce. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying MapReduce*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Demonstrate the basic concepts of Hive. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate the basic concepts of Hive*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Apply Hive Query Language Statements. (3 marks)

Answer: Begin with the standard definition or identity of *Apply Hive Query Language Statements*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Make use of Hive User Defined Function. (3 marks)

Answer: Begin with the standard definition or identity of *Make use of Hive User Defined Function*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Make use of the basic concepts of Pig. (3 marks)

Answer: Begin with the standard definition or identity of *Make use of the basic concepts of Pig*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Summarize the Characteristics of Big Data**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Analytics and the major challenges faced 3 Understanding by Big Data.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the fundamentals of Hadoop**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Demonstrate the basic concepts of Hive**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://hadoop.apache.org/
- https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ https://hive.apache.org/
- https://www.tutorialspoint.com/hadoop/index.htm https://pig.apache.org/

IN-PAGE SEARCH

Search this lesson